Shivendra Agrawal

Curriculum Vitae

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Summary - I completed my Masters in CS from CU Boulder and have re-joined the school as a **Ph.D student from Fall 2019**. Prior to this, I have experience at Noodle.ai and Bosch as a data scientist and as a co-founder of a start-up. I am broadly interested in Human Robotics Interaction, Assistive Technology, Computer Vision, 3D Scene Understanding, Reinforcement Learning, and Explainable AI. **My work involves creating efficient, safe and explanable Human Robot Interaction (HRI) focused on helping visually impaired people.**.

Education

2019– **University of Colorado, Boulder**, *Ph.D. Cohort*, *Computer Science*.

2017–2019 **University of Colorado, Boulder**, *GPA: 4.0/4.0*, Masters of Science, *Computer Science*.

- Teaching Assistant - Awarded **outstanding TA** for **Algorithms** (2017-2018, 2018-2019)
- Instructor - **Teaching more than a 100 students** for undergraduate algorithms in Fall 2019.

2009–2014 **Indian Institute of Technology, Kharagpur**, *Dual Degree*, *Mathematics Computing*.

Papers

September 2018 **Explanation-based Reward Coaching to Improve Human Performance via Reinforcement Learning**, *Best paper nominee, HRI 2019*, CU Boulder.
We introduced a new framework to infer the human collaborators' mental view of the optimum state, predict future failure based on the inferred human mental model and provide interpretable feedbacks to help the human in a shared task. (Link to the paper)

Projects

Sep 2019 - ongoing **Manipulation assistance for blind using a robot arm**, Prof. Brad Hayes, CU Boulder.
Developing algorithms for various modalities of assistance for table top manipulation. Modalities include multiple levels of verbal or physical guidance. The task involves risk due to dangerous objects on the table. We will carry out a human subject study to estimate factors like efficiency and ease of our modalities.

Feb 2019 - Apr 2019 **Robot guide dog for visually impaired**, Prof. Brad Hayes, CU Boulder.
In a first step towards my Ph.D. thesis we (in a team of 3) tried to emulate a guide dog in indoor scenario. We proposed a solution based on mobile robot platform. The robot would provide navigation assistance and scene explanation using a physical lease feedback and natural language interface. (Video demo)

Feb 2019 - Apr 2019 **Visual memory with sparse demonstrations**, *Prof. Chris Heckman*, CU Boulder.
Explored learning a path information using visual cues extracted via a CNN so that the robot can re-trace the path without the 3D map of the environment and with very less demonstrations of the path.
Showed that a vanilla LSTM is still insufficient to generate crucial policies just from a few demonstration and may need attention going forward. (Link to project)

August 2018 - December 2018 **An ARKit app to help people with learning disabilities**, *Virtual Reality*, CU Boulder.
We made an IOS app with a python ML backend that parsed text from images and represented it in a more intuitive form that enabled learning. We also created a text editor which would help a person with dyslexia to just focus on some text.
My role - I designed the python backend, the IOS app, the text editor, came up with a way to translate from 2D text image coordinates to 3D world coordinates.(Link to project)

April 2018 – May 2018 **Logo Generation using GAN**, *Human Centered Machine Learning*, CU Boulder.
We (2 membered team) created a GAN based Logo generator to help generate original and unique initial logo designs. Tried out various variations like WGAN and ACGAN. (Link to report)

May 2018 **Twitter Trend Analysis**, *Big-Data Architecture*, CU Boulder.
We (5 membered team) developed a scalable web app using Kafka, Cassandra, and Spark which could classify upto 6k tweets per minute in real-time. *My role* - project conceptualization, trend similarity, leaflet map and front-end development.(Video demo)

Oct 2017 – Dec 2017 **Image Depth Map Estimation**, *Machine Learning*, CU Boulder.
Compared various CNN based architectures including residual layers and fully convolutional networks using Keras. Demonstrated an application of the predicted depth to create *"Portrait Mode"*. (Report link)

Oct 2017 – Dec 2017 **Motion Planning with RRT**, *Algorithms*, CU Boulder.
Created a Javascript based GUI implementation of RRT based algorithm for path finding. Compared this with vanilla A* varying multiple settings including sampling percentage, complexity of obstacles and tree expansion distance. (Link to report)

Previous Work Experience

Sep 2016 – May 2017 **Associate Data Scientist**, Noodle.ai, Bangalore, India.

- ***Exploratory Data Analysis (EDA) tool*** - Developed a Django app to perform preliminary data analysis. The EDA tool automatically munges data to plot the features, performs statistical tests, find interesting inferences and summarizes the result.
- ***Demand forecast demo tool*** - Developed a demo tool which uses dummy data and shows micro-demand (store-level) predictions focusing FMCG companies. This tool was used a multiple times and proved to be more effective than a slideshow.

Mar 2016 – Aug 2016 **Co-Founder**, Progyrus, Bangalore, India.

- Progyrus provided (Uber-like) large scaled human powered translations to help brands go vernacular. We worked with 6 clients to pilot regional content with 50 translators on the platform
- I contributed in conceptualizing the idea and launched the MVP product. Designed the order management and task management systems. Developed dashboards using AngularJS. Used Django REST framework + PostgreSQL and developed APIs to automate the translation workflow and integration at client side.

June 2014 – Feb 2016 **Senior Engineer (Analytics)**, Robert Bosch GmbH, Bangalore, India.

- **Prototype optimization for an automobile manufacturer** - Developed code to computationally form feasible prototypes (satisfying all boolean formulas) to reduce the number of actual test car productions. Devised and parallelized the algorithms that modeled the problem into a maximal clique finding problem.
- **Ticket Analysis** - Analyzed texts in complain tickets and came up with a novel hierarchical clustering method to identify common problems and sub-problems and made D3.js visualization and a GUI tool.

Competitions

2015 **D4D Data Challenge**, Orange Telecom.
Developed a system using markov clustering to analyze population movement and commute clusters for Senegal using cell phone data. We presented a poster on our analysis at NetMob 2015 in MIT, Cambridge.

Skills & Abilities

Languages Python, Javascript, C++

Libraries AngularJS, Django REST, Flask, Keras, Scikit-Learn, Selenium, Elasticsearch, PyTorch

Interests

YouTube I have started a **YouTube channel** to explain some complicated stuff from algorithms, robotics, AI etc and have had a good start with my first video. Checkout my channel at - (Informal-CS)

Food I like cooking and exploring various cuisines and have 6 million total views on Google Maps.

Biking I am a beginner but this is one of the best part of the day for me.